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Signals and Systems

Complete Lecture Notes

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I. Signals

1. **Signal Classification**

- Time: Continuous-time vs. Discrete-time
- Continuous-time: Analog Signal
- Discrete-time: Digital (Quantized) Signal

2. **Signal Decomposition**

- ① Convergence Property (Transforms):
- Continuous-time non-periodic: Fourier Transform (FT)

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

- Continuous-time periodic: Fourier Series (FS)

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_0 t}$$

- Discrete-time non-periodic: Discrete-Time Fourier Transform (DTFT)

$$f[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(e^{j\omega}) e^{j\omega n} d\omega$$

- Discrete-time periodic: Discrete Fourier Series (DFS)

$$f[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn}$$

- ② Orthogonality:
- Inner Product (Continuous): $\langle f_1(t), f_2(t) \rangle = \int_{-\infty}^{\infty} f_1(t) f_2^*(t) dt$

- Inner Product (Discrete): $\langle f_1[n], f_2[n] \rangle = \sum_{n=-\infty}^{\infty} f_1[n] f_2^*[n]$
- Orthogonal condition: $\langle \phi_m, \phi_n \rangle = \lambda_n \delta_{mn}$

Energy Signals:

- Energy: $E_f = \int_{-\infty}^{\infty} |f(t)|^2 dt < \infty$
- Autocorrelation: $R_{ff}(\tau) = \int_{-\infty}^{\infty} f(t) f^*(t-\tau) dt$
- Total Energy: $E = R_{ff}(0)$

Power Signals:

- Average Power: $P = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt < \infty$
- Autocorrelation: $R_{ff}(\tau) = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} f(t) f^*(t-\tau) dt$
- Average Power: $P = R_{ff}(0)$

II. Systems

1. **System Classification**
   - Linear vs. Nonlinear (Superposition & Homogeneity)
   - Time-Invariant vs. Time-Varying:

$$x(t - t_0) \to y(t - t_0)$$

   - Causal vs. Non-causal (Output depends only on present and past inputs)
   - Stable vs. Unstable (BIBO Stability)
   - Memory vs. Memoryless
2. **System Representations**
   - Block Diagrams: Integrators, adders, multipliers
   - Differential Equations (Continuous) & Difference Equations (Discrete)

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Linear Time-Invariant (LTI) Systems

1. **Continuous-Time LTI Systems: Convolution Integral**

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

where $h(t)$ is the impulse response of the system.

2. **Properties of Convolution:**

- Commutative: $x(t) * h(t) = h(t) * x(t)$
- Associative: $[x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)]$
- Distributive: $x(t) * [h_1(t) + h_2(t)] = x(t) * h_1(t) + x(t) * h_2(t)$

3. **Impulse Response of Interconnected Systems:**

- Parallel Connection: $h(t) = h_1(t) + h_2(t)$
- Cascade Connection: $h(t) = h_1(t) * h_2(t)$

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III. Time Domain Analysis of LTI Systems

1. **System Response Decomposition:**

- Total Response = Zero-Input Response ($y_{zi}(t)$) + Zero-State Response ($y_{zs}(t)$)
- Total Response = Natural Response + Forced Response

2. **System Properties from Impulse Response $h(t)$:**

- Causality: $h(t) = 0$ for $t < 0$
- BIBO Stability: $\int_{-\infty}^{\infty} |h(t)| dt < \infty$

3. **Impulse Response Matching Method:**

Used to find initial conditions at $t = 0^+$ from conditions at $t = 0^-$ when impulse delta inputs are present.

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IV. Laplace Transform and s-Domain Analysis

1. **Bilateral Laplace Transform:**

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt, \quad s = \sigma + j\omega$$

2. **Region of Convergence (ROC):**

- Determined by the real part $\sigma = \text{Re}(s)$ where the integral converges.

- For right-sided signals, ROC is $\text{Re}(s) > \sigma_{\max}$.
- For left-sided signals, ROC is $\text{Re}(s) < \sigma_{\min}$.
- For double-sided signals, ROC is a strip $\sigma_1 < \text{Re}(s) < \sigma_2$.

3. **Partial Fraction Expansion (PFE) for Inverse Laplace:**

- Simple real poles: $H(s) = \sum_{i=1}^{n} \frac{A_i}{s-p_i} \leftrightarrow h(t) = \sum_{i=1}^{n} A_i e^{p_i t} u(t)$
- Complex conjugate poles: $H(s) = \frac{As+B}{(s+\alpha)^2+\beta^2} \leftrightarrow h(t) = Ce^{-\alpha t} \cos(\beta t + \theta) u(t)$

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Laplace Transform Properties

Property	Time Domain $x(t)$	s-Domain $X(s)$	ROC
Linearity	$a_1 x_1(t) + a_2 x_2(t)$	$a_1 X_1(s) + a_2 X_2(s)$	Contains $R_1 \cap R_2$
Time Shifting	$x(t - t_0)$	$e^{-st_0} X(s)$	Same as R
Shifting in s	$e^{s_0 t} x(t)$	$X(s - s_0)$	Shifted by $\text{Re}(s_0)$
Scaling	$x(at)$	$\frac{1}{\lvert a \rvert} X(\frac{s}{a})$	Scaled by a
Differentiation in Time	$\frac{dx(t)}{dt}$	$sX(s) - x(0^-)$	Contains R
Integration in Time	$\int_{-\infty}^{t} x(\tau) d\tau$	$\frac{X(s)}{s}$	Contains $R \cap \{\text{Re}(s) > 0\}$
Convolution	$x_1(t) * x_2(t)$	$X_1(s) X_2(s)$	Contains $R_1 \cap R_2$

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V. Fourier Analysis of Continuous-Time Signals

1. **Continuous-Time Fourier Transform (CTFT):**

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$$

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}d\omega$$

2. **Fourier Transform of LTI System Output:**

$$Y(j\omega) = X(j\omega)H(j\omega)$$

where $H(j\omega)$ is the Frequency Response:

$$H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t}dt$$

3. **LTI System Properties in Frequency Domain:**

- Distortionless Transmission: $y(t) = Kx(t - t_0) \leftrightarrow H(j\omega) = Ke^{-j\omega t_0}$
- Ideal Low-Pass Filter: $H(j\omega) = 1$ for $|\omega| \leq \omega_c$, and 0 otherwise.

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Fourier Transform of Periodic Signals

If $f(t)$ is periodic with period T (fundamental frequency $\omega_0 = \frac{2\pi}{T}$):

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_0 t}$$

$$F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_0)$$

where the Fourier coefficients are:

$$F_n = \frac{1}{T}\int_{-T/2}^{T/2} f(t)e^{-jn\omega_0 t}dt$$

Autocorrelation function for periodic power signals:

$$R_f(\tau) = \frac{1}{T}\int_{-T/2}^{T/2} f(t)f^*(t-\tau)dt = \sum_{n=-\infty}^{\infty} |F_n|^2 e^{jn\omega_0 \tau}$$

$$S_f(\omega) = 2\pi \sum_{n=-\infty}^{\infty} |F_n|^2 \delta(\omega - n\omega_0)$$

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VI. Discrete-Time Signals and Systems (Time Domain)

1. **Representation of Discrete Signals:**

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

2. **Discrete Convolution Sum (LTI Systems):**

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

3. **Solving Linear Difference Equations:**
   - Homogeneous solution $y_h[n]$ (roots of characteristic equation).
   - Particular solution $y_p[n]$ (matching the form of input $x[n]$).
   - Total solution: $y[n] = y_h[n] + y_p[n]$.

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VII. The Hilbert Transform

The Hilbert Transform of $x(t)$, denoted $\hat{x}(t)$, is the convolution of $x(t)$ with $\frac{1}{\pi t}$:

$$\hat{x}(t) = x(t) * \frac{1}{\pi t} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x(\tau)}{t-\tau} d\tau$$

In the frequency domain, the frequency response is:

$$\hat{X}(j\omega) = X(j\omega)H(j\omega) = X(j\omega) \cdot (-j\text{sgn}(\omega))$$

For an analytic signal $z(t) = x(t) + j\hat{x}(t)$:

$$Z(j\omega) = \begin{cases} 2X(j\omega), & \omega > 0 \\ X(j\omega), & \omega = 0 \\ 0, & \omega < 0 \end{cases}$$

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VIII. The z-Transform

1. **Bilateral z-Transform:**

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

2. **Region of Convergence (ROC):**
- Right-sided signal: ROC is $|z| > r_{\max}$
- Left-sided signal: ROC is $|z| < r_{\min}$
- Two-sided signal: ROC is a ring $r_i < |z| < r_o$
3. **Stability & Causality in z-Domain:**
- An LTI system is **stable** if the ROC of the system function $H(z)$ contains the unit circle $|z| = 1$.

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

- An LTI system is **causal** if the ROC is the region outside a circle containing all poles.

$$\text{ROC: } |z| > r_{\max}$$

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z-Transform Properties

1. **Linearity:** $a_1 x_1[n] + a_2 x_2[n] \leftrightarrow a_1 X_1(z) + a_2 X_2(z)$
2. **Time Shifting:** $x[n - n_0] \leftrightarrow z^{-n_0} X(z)$
3. **Scaling in z-Domain:** $a^n x[n] \leftrightarrow X(a^{-1} z)$

4. **Time Reversal:** $x[-n] \leftrightarrow X(z^{-1})$
5. **Conjugation:** $x^*[n] \leftrightarrow X^*(z^*)$
6. **Convolution:** $x_1[n] * x_2[n] \leftrightarrow X_1(z)X_2(z)$
7. **Differentiation in z-Domain:** $nx[n] \leftrightarrow -z\frac{dX(z)}{dz}$
8. **Initial Value Theorem:** If $x[n]$ is causal, $x[0] = \lim_{z\to\infty} X(z)$
9. **Final Value Theorem:** If $x[n]$ is causal, $x[\infty] = \lim_{z\to 1}(z-1)X(z)$

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IX. Discrete-Time Fourier Transform (DTFT)

1. **Definition of DTFT:**

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\omega}$$

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega$$

2. **Relation to z-Transform:**
   If the ROC of $X(z)$ contains the unit circle $|z| = 1$, the DTFT is the z-transform evaluated on the unit circle:

$$X(e^{j\omega}) = X(z)\Big|_{z=e^{j\omega}}$$

3. **Frequency Response of Discrete LTI Systems:**

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

   where $H(e^{j\omega})$ is the system's frequency response.

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DTFT of Periodic Sequences & DFS

1. **Discrete Fourier Series (DFS) for Periodic Sequences:**

For periodic sequence $x[n]$ with period N:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{j\frac{2\pi}{N}kn}$$

$$X_k = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

2. **DTFT of a Periodic Sequence:**

$$X(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} X_k \delta\left(\omega - \frac{2\pi k}{N}\right)$$

3. **Sampling the DTFT:**
   For a finite-length sequence $x[n]$ of length N_0, if we sample its DTFT $X(e^{j\omega})$ at N points ($N \geq N_0$), the samples are the DFT coefficients $X[k]$:

$$X[k] = X(e^{j\omega})\Big|_{\omega=\frac{2\pi k}{N}} = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N}n}$$

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X. Sampling and Reconstruction

1. **Impulse Train Sampling:**
   Sampling a continuous signal $x_a(t)$ with interval T:

$$x_s(t) = x_a(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} x_a(nT)\delta(t - nT)$$

2. **Spectrum of the Sampled Signal (Nyquist Shannon):**

$$X_s(j\omega) = \frac{1}{T} \sum_{m=-\infty}^{\infty} X_a\left(j\left(\omega - m\frac{2\pi}{T}\right)\right)$$

3. **Nyquist Sampling Theorem:**
   To reconstruct $x_a(t)$ without aliasing, the sampling frequency $\omega_s = \frac{2\pi}{T}$ must satisfy:

$$\omega_s > 2\omega_{\max}$$

where $\omega_{\max}$ is the maximum frequency component of $x_a(t)$.

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XI. The Discrete Fourier Transform (DFT) & FFT

1. **Discrete Fourier Transform (DFT):**

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad W_N = e^{-j\frac{2\pi}{N}}, \quad k = 0, 1, \ldots, N-1$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, \quad n = 0, 1, \ldots, N-1$$

2. **Decimation-in-Time Fast Fourier Transform (DIT-FFT):**
   Splitting $x[n]$ of size N into even and odd indices:

$$X[k] = G[k] + W_N^k H[k]$$

$$X[k + N/2] = G[k] - W_N^k H[k]$$

   where $G[k]$ and $H[k]$ are the $N/2$-point DFTs of the even and odd sequences.

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XII. Multirate Signal Processing

1. **Decimation (Downsampling by M):**
   Time Domain relation:

$$y[n] = v[nM]$$

   Frequency Domain relation:

$$Y(e^{j\omega}) = \frac{1}{M} \sum_{i=0}^{M-1} V\left(e^{j\frac{\omega - 2\pi i}{M}}\right)$$

2. **Interpolation (Upsampling by L):**

   Time Domain relation:

$$w[n] = \begin{cases} x[n/L], & n = 0, \pm L, \pm 2L, \ldots \\ 0, & \text{otherwise} \end{cases}$$

   Frequency Domain relation:

$$W(e^{j\omega}) = X(e^{j\omega L})$$

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XIII. Filter Structures and Design

1. **Analog Filter Transfer Function:**

$$H_a(s) = \frac{V_o(s)}{E_s(s)}$$

2. **Lossless Network Power transmission:**

   For a lossless network terminated in resistances, the transmission coefficient is $|s_{12}(s)|^2$:

$$|H_a(j\Omega)|^2 = \frac{R_L}{4R_s}\left|\frac{V_o(j\Omega)}{E_s(j\Omega)}\right|^2$$

   Reflection coefficient function:

$$|\rho(s)|^2 = 1 - 4\frac{R_s}{R_L}|H_a(s)|^2$$

3. **Digital Filter Realization (Difference Equations):**

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

$$y[n] = \sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k]$$

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Butterworth Filter Design

1. **Magnitude Squared Response:**

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

2. **Pole Locations:**

$$s_k = \Omega_c e^{j\left[\frac{\pi}{2} + \frac{2k-1}{2N}\pi\right]}, \quad k = 1, 2, \ldots, 2N$$

3. **Stable Transfer Function $H_a(s)$:**
Formed by taking only the N poles in the left half-plane (LHP):

$$H_a(s) = \frac{\Omega_c^N}{\prod_{k \in \text{LHP}} (s - s_k)} = \frac{\Omega_c^N}{B_N(s')}$$

where $s' = s/\Omega_c$ is the normalized Laplace variable, and $B_N(s')$ is the normalized Butterworth polynomial.

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XIV. Infinite Impulse Response (IIR) Digital Filter Design

1. **Impulse Invariance Method:**
- Relationship: $h[n] = T_d h_a(nT_d)$
- Mapping in s to z domain:

$$\frac{A}{s - s_0} \Rightarrow \frac{AT_d}{1 - e^{s_0 T_d} z^{-1}}$$

- Disadvantage: Frequency domain aliasing can occur.
2. **Bilinear Transform Method:**
- Mapping relation:

$$s = \frac{2}{T_d} \frac{1 - z^{-1}}{1 + z^{-1}}$$

- Frequency warping relation:

$$\Omega = \frac{2}{T_d} \tan\left(\frac{\omega}{2}\right)$$

- Advantage: No aliasing, but frequency mapping is non-linear (warped).

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XV. Finite Word Length Effects in Digital Filters

1. **Quantization Noise Models:**
- Rounding (Interval $[-q/2, q/2]$): Mean 0, Variance $\sigma_e^2 = \frac{q^2}{12}$
- Truncation (Interval $[-q, 0]$): Mean $-q/2$, Variance $\sigma_e^2 = \frac{q^2}{12}$
- Quantization step size: $q = 2^{-B}$ where B is the number of bits.
2. **Output Noise Variance:**
   Quantization noise passing through LTI system:

$$\sigma_{yo}^2 = \sigma_e^2 \sum_{n=-\infty}^{\infty} |h[n]|^2 = \sigma_e^2 \frac{1}{2\pi} \int_{-\pi}^{\pi} |H(e^{j\omega})|^2 d\omega$$

3. **Scaling to Prevent Overflow:**
   To ensure node variables do not overflow $|w[n]| \leq 1$ under input $|x[n]| \leq 1$, we scale the input by a factor β:

$$\beta \leq \frac{1}{\sum_{n=-\infty}^{\infty} |f[n]|} \quad (L_1 \text{ bound})$$

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XVI. Linear Phase FIR Filter Design

1. **Conditions for Linear Phase:**
   Symmetry condition: $h[n] = h[N-1-n]$ (Type 1 & 2)

   Anti-symmetry condition: $h[n] = -h[N-1-n]$ (Type 3 & 4)

2. **Frequency Response $H(e^{j\omega})$ Types:**

- Type 1 ($h[n]$ symmetric, N odd): $H(e^{j\omega}) = e^{-j\omega(N-1)/2} \sum_{n=0}^{(N-1)/2} a[n] \cos(\omega n)$
- Type 2 ($h[n]$ symmetric, N even): $H(e^{j\omega}) = e^{-j\omega(N-1)/2} \sum_{n=1}^{N/2} b[n] \cos\left(\omega\left(n - \frac{1}{2}\right)\right)$

3. **Digital Filter Frequency Transformation:**

   Frequency mapping on unit circle to transform a lowpass filter $H(Z)$ to other types:

$$Z^{-1} = G(z^{-1}) = \pm \frac{z^{-1} - \alpha}{1 - \alpha z^{-1}}$$

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XVII. State-Space Representation of Systems

1. **Continuous-Time State-Space Equations:**

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{B}\boldsymbol{u}(t)$$
$$\boldsymbol{y}(t) = \boldsymbol{C}\boldsymbol{x}(t) + \boldsymbol{D}\boldsymbol{u}(t)$$

- Time-domain Solution:

$$\boldsymbol{x}(t) = e^{\boldsymbol{A}t}\boldsymbol{x}(0) + \int_0^t e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}\boldsymbol{u}(\tau)d\tau$$

- Laplace Domain Solution:

$$\boldsymbol{X}(s) = (s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{x}(0) + (s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{U}(s)$$
$$\boldsymbol{Y}(s) = \boldsymbol{C}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{x}(0) + [\boldsymbol{C}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B} + \boldsymbol{D}]\boldsymbol{U}(s)$$

2. **Discrete-Time State-Space Equations:**

$$\boldsymbol{x}[n+1] = \boldsymbol{A}\boldsymbol{x}[n] + \boldsymbol{B}\boldsymbol{u}[n]$$
$$\boldsymbol{y}[n] = \boldsymbol{C}\boldsymbol{x}[n] + \boldsymbol{D}\boldsymbol{u}[n]$$

- Time-domain Solution:

$$\boldsymbol{x}[n] = \boldsymbol{A}^n\boldsymbol{x}[0] + \sum_{k=0}^{n-1} \boldsymbol{A}^{n-1-k}\boldsymbol{B}\boldsymbol{u}[k]$$

- z-Domain Solution:

$$\boldsymbol{X}(z) = z(z\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{x}[0] + (z\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{U}(z)$$
$$\boldsymbol{Y}(z) = z\boldsymbol{C}(z\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{x}[0] + [\boldsymbol{C}(z\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B} + \boldsymbol{D}]\boldsymbol{U}(z)$$

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State-Space Coordinate Transformation

1. **Linear State Transformation:**
   Let $\boldsymbol{x}'[n] = \boldsymbol{P}\boldsymbol{x}[n]$, where $\boldsymbol{P}$ is a non-singular transformation matrix. The new state equations are:

$$\boldsymbol{A}' = \boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{-1}, \quad \boldsymbol{B}' = \boldsymbol{P}\boldsymbol{B}, \quad \boldsymbol{C}' = \boldsymbol{C}\boldsymbol{P}^{-1}, \quad \boldsymbol{D}' = \boldsymbol{D}$$

2. **Controllability and Observability:**
- An LTI system is **controllable** if the controllability matrix $\boldsymbol{Q}_c$ has full rank N:

$$\boldsymbol{Q}_c = [\boldsymbol{B}, \boldsymbol{A}\boldsymbol{B}, \boldsymbol{A}^2\boldsymbol{B}, \ldots, \boldsymbol{A}^{N-1}\boldsymbol{B}]$$

- An LTI system is **observable** if the observability matrix $\boldsymbol{Q}_o$ has full rank N:

$$\boldsymbol{Q}_o = \begin{bmatrix} \boldsymbol{C} \\ \boldsymbol{C}\boldsymbol{A} \\ \boldsymbol{C}\boldsymbol{A}^2 \\ \vdots \\ \boldsymbol{C}\boldsymbol{A}^{N-1} \end{bmatrix}$$
